



A study of direct welding of dissimilar materials of borosilicate glass and 304 stainless steel using a green picosecond laser

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ABSTRACT

Welding borosilicate glass to stainless steel is challenged by their vastly different thermophysical properties, which typically induces stress and cracking. Ultrafast lasers overcome this by utilizing non-thermal, nonlinear absorption to create joints with minimal heat input. However, the nanoscale mechanisms governing interface formation are still unclear. In this study, borosilicate glass and stainless steel were successfully joined using a green picosecond laser. The effects of key parameters including laser power, repetition frequency, and pressing force on the weld quality were systematically investigated. Advanced characterization via SEM and TEM revealed that the interfacial microstructure consists of laser-induced periodic surface structures (LIPSS) and blocky glass, with the transition zone containing amorphous, crystalline, and mixed phases. Elemental migration was observed to mitigate composition segregation, thereby enhancing bonding. Although microcracks formed due to thermally induced stresses from material property differences, the joints achieved an average load capacity of 67.14 N under optimal parameters. The improved mechanical performance is primarily attributed to the mechanical interlocking from serrated interface structures and the reduced microstructural gradients across the reaction layers. This work provides significant insights into the interfacial mechanisms of glass-metal welding and demonstrates the potential of picosecond laser processing for high-performance dissimilar material joining in the field of manufacturing.

1. Introduction

Glass is a representative functional material with good transmittance, high wear resistance, corrosion resistance and temperature resistance. However, the mechanics performance of glass is rather poor on account of its great brittleness and low impact toughness, which greatly restricts its application in engineering structures [1–3]. The joining of glass and metal with high plastic toughness and strong impact resistance can greatly enhance the overall mechanical property. Consequently, there has been considerable interest in the fields of optics, electronics, communications, aerospace and biomedical sciences in possible benefits arising from the bonding of glass and metal for industrial applications [4–7].

Welding is an important means to realize the connection of dissimilar materials in the manufacturing field. In order to achieve a good joint quality, multiple welding methods have been experimentally explored

by some researchers to join dissimilar materials of glass and metal. Weber and Rettenmayr [8] conducted the brazing of silica glass to 316L stainless steel, and successfully obtained a tight joint with several alloy compositions. Hu et al. [9] presented a novel glass-metal coupling technique combining anodic bonding and Sn–9Zn soldering, and revealed a four-stage interface evolution where optimal shear strength correlates with controlled intermetallic growth. Ren et al. [10] studied the effects of laser soldering parameters on the mechanical property of glass-metal joints, and determined that improved bonding strength depended mainly on the increase of laser power and irradiation time. However, excessive heat input resulted in glass fragmentation. Harinarayana and Shin [11] performed continuous wave fiber laser transmission welding of borosilicate glass and stainless steel 304 L, and proved that keeping interface temperature close to the softening point of the glass helped to achieve optimal welds. Moreover, it has been proved that exceeding this threshold would result in the occurrence of defects

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Table 1
The chemical composition of the base materials (mass fraction, %).

Element	C	Si	Mn	Cu	Ni	Cr	S	P	Fe
SUS304	0.08	0.06	1.28	1.67	9.30	17.61	0.001	0.019	69.98
Element	Si	B	Al	Na	K	O			
KG-3	39.37	3.59	1.66	3.16	1.75	50.46			

such as bubbles, cracks and fractures. Zhang et al. [12] achieved hermetic sealing of glass to an aluminum alloy using millisecond pulsed laser welding. A micro-arc oxidation coating acted as an energy-absorbing buffer layer, enabling the formation of a mixed mosaic weld structure. Although these welding methods can achieve the bonding of glass and metal, it is very difficult to obtain good joint with good performance because of their transparency, hardness, fragileness, high glass transition temperature and great difference in thermal expansion coefficient. Therefore, the bonding of glass and metal remains a challenge.

In order to further improve joint quality, nanosecond lasers were also used for precise welding of glass and metal dissimilar materials because of their advantages of short pulse duration, high peak powers and high repetition rate. Morawska et al. [13] developed a picosecond laser welding process for stainless-steel and quartz. Their optimized technique produced strong, reliable bonds that survived rigorous thermal and vibration tests, demonstrating high industrial viability for optical assemblies. Wang et al. [14] achieved direct welding of ZnS crystals to Ti–6Al–4V alloy using a picosecond laser. The joint attained a record-high shear strength of 67.7 MPa and significantly reduced residual stress, demonstrating great potential for optoelectronic and aerospace applications. Yang et al. [15] introduced the concept of a welding threshold for picosecond laser welding of quartz glass and 304 stainless steel. They identified three welding behaviors based on pulse energy density and effective pulse number, providing guidance for parameter selection in ultrafast laser welding. Huo et al. [16] explored

infrared nanosecond laser welding of silica glass and 304 stainless steel, and confirmed that the element diffusion (such as Fe, Cr and Si) was the decisive factors in forming the welds during the welding process. Li et al. [17] used nanosecond laser to weld glass and metal by means of interstitial fluid surface tension without clampers, and proposed a pre-welding method to broaden available range of process parameters. Huo et al. [18] introduced chromium interlayer to join silica glass and stainless steel during nanosecond laser welding, and found that the improvement of joint strength attributed to the increase of Fe–O–Si and Cr–Si bonds and the decrease of microcracks in the welds. Nguyen et al. [19,20] focused on nanosecond pulsed laser transmission welding of glass and copper, and ascertained that high weld strength was closely related to the correlation between weld zone area, eutectic Cu^{x+} -O-Si compounds and uniformity of re-solidified glass region. Feng et al. [21] carried out direct joining of quartz glass and copper by nanosecond laser, and concluded that the formation of spindle microstructures depended mainly on the recoil pressure and the Marangoni convection during nanosecond laser welding. The joint strength was improved by the coupling effect of the metallurgical bonding and mechanical interlocking at the interface. Compared with conventional welding methods, nanosecond pulsed laser transmission welding can improve the welding quality of glass and metal [22,23]. However, it is very difficult to eradicate the defects such as cracks or voids in joints welded with nanosecond lasers. In order to effectively suppress welding defects in glass-metal joints, the interaction between laser and material requires precision control during laser welding. Ultrafast laser has the

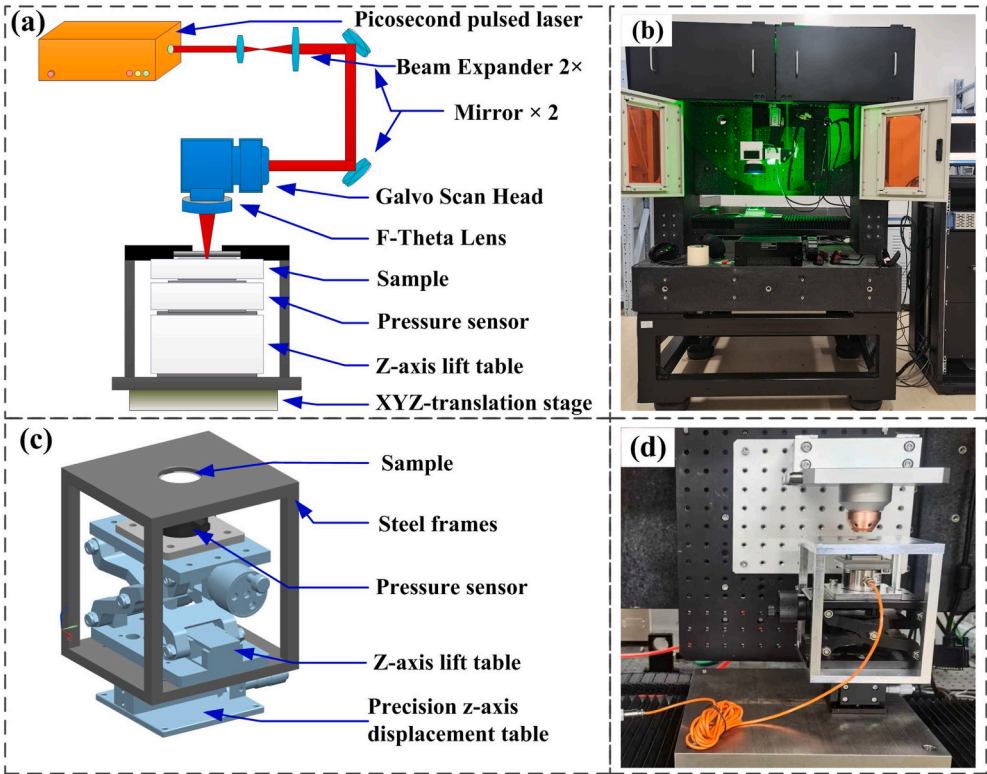


Fig. 1. Schematic and physical diagrams of the ultrafast pulsed laser welding device: (a) schematic and (b) physical diagram of the welding platform; (c) schematic and (d) physical diagram of the laser welding fixture.

Table 2

Process variables of picosecond laser welding.

Equipment	Variables	Units	Levels
Green picosecond laser	Pressing force	N	0, 400
	Pulse laser power	W	12, 16, 20
	Repetition frequency	kHz	600, 800, 1000
	Welding speed	mm/s	50
	Number of welds	/	10

characteristics of ultrashort pulse width, ultrahigh peak power and ultrahigh repetition rate [24,25], which is considered to be an ideal means of joining dissimilar materials glass and metal. Some research showed that voids or cracks in the welds were successfully limited on account of the inhibiting effect of heat and the precise control of the welded region during femtosecond laser welding of copper and glass [26,27]. Therefore, ultrafast laser welding is a promising welding technology that is expected to achieve high-quality connections between metal and glass. Current research on metal and glass welding by ultrafast laser has mainly concentrated upon the stage of feasibility study, while few studies have been reported on interfacial microstructure and the bonding mechanism of metal and glass using ultrafast laser welding.

In this work, direct welding was conducted on dissimilar materials of borosilicate glass and 304 stainless steel using 170 mm-focus green picosecond laser. The interfacial microstructure and mechanical property were carefully investigated using scanning electron microscope (SEM), energy dispersive spectroscopy (EDS) and transmission electron microscope (TEM). The mechanical property of welded joints was tested on versatile testing machine. Moreover, the welding mechanism of borosilicate glass and stainless steel was also explored based on microstructural characteristics at the interface and performance evaluation.

2. Experimental materials and methods

The materials used in the experiments were borosilicate glass KG-3 with a size of 16 mm × 10 mm × 1 mm and SUS304 stainless steel with a size of φ30 mm × 8 mm. The chemical compositions of KG-3 glass and SUS304 stainless steel are listed in Table 1.

Prior to experimentation, the pre-welded surface of SUS304 metal was polished using alumina/silica suspension containing abrasive particles <0.06 μm. In order to obtain the good welding quality, the surface roughness of SUS304 and borosilicate glass should be controlled to less than Ra 0.03 μm and Ra 0.01 μm, respectively. Moreover, the gap between the two materials must be also controlled to less than 133 nm, which achieved an optical contact condition [28].

The schematic diagram of ultrafast pulsed laser welding device is shown in Fig. 1. The welding experimental devices consisted mainly of

laser source, focusing system, clamping tool and moving platform. The laser source is a 532 nm ps pulsed solid state laser (Huaray, pine2-532-60) with a pulse width of 10 ps, a maximum laser power of 60 W, and a repetition rate of 400–1000 kHz. The focusing system is evenoptics (Jenoptik, JENar) with a focal length of 170 mm and a minimum spot diameter of 30 μm.

The clamping tool was designed for the size of the sample to be welded. It consisted mainly of steel frames, pressure sensor, Z-axis lift table and precision Z-axis displacement table. The three-dimensional mobile device was used for precisely adjusting the position of welded samples and determining the focal plane height. Before laser welding, the glass was firstly placed on SUS304 before assembling into the clamping tool. A desired clamping force between the SUS304 and the glass was successfully obtained by moving the weldment to contact the fused silica installed on the top of the fixture. During the clamping process, the imposed force can be measured by the pressure sensor installed beneath the weldment.

The main parameters were tested at different levels in picosecond laser welding, including laser power, welding speed and the repetition frequency. The main process variables were given in Table 2.

In order to investigate the influence of welding parameters on the weld quality, metallographic samples from the cross sections of the joint were prepared. Due to the non-conductive nature of the glass and the small size of the weld seam, the samples were precisely sectioned using a diamond wire saw and subsequently cold-mounted with resin. The mounted specimens were then ground and polished to obtain standard metallographic cross-sections of the weld region.

After being sputter-coated with a thin Au layer to ensure adequate electrical conductivity, the samples were first conducted to examine microstructural features near the SUS304 interface using backscattered electron (BSE) imaging mode in scanning electron microscopy (ZEISS, GeminiSEM). The SEM was operated at an accelerating voltage of 15 kV and a working distance of 8.5 mm. The origin and evolution of the morphology were further analyzed by energy dispersive spectroscopy (Oxford, Aztec) for elemental mapping. For nanoscale analysis, cross-sectional TEM lamellae were prepared using a focused ion beam (FIB) system and examined using a transmission electron microscope (ThermoFisher, F200x) operated at 200 kV. The mechanical performance of the welded joints was evaluated using an electrodynamic dynamic and static fatigue testing machine (Instron, E1000). Shear tests were conducted at a constant displacement rate of 0.3 mm/min until failure. Shear tests were repeated three times for joints fabricated under identical parameters, and the average value of the maximum load was used to evaluate the joint strength. Fig. 2 illustrates the shear test configuration, including loading direction and fixture geometry.

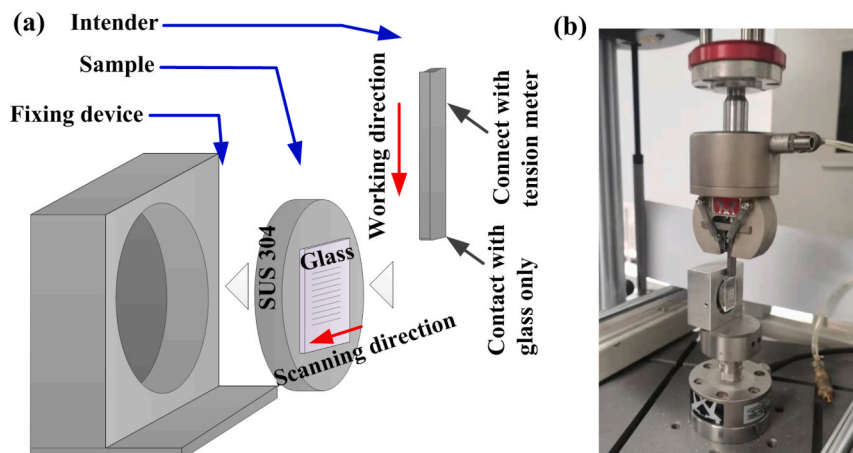


Fig. 2. Schematic and physical diagrams of the clamping device for testing joint shear strength: (a) schematic and (b) physical diagram of the device.

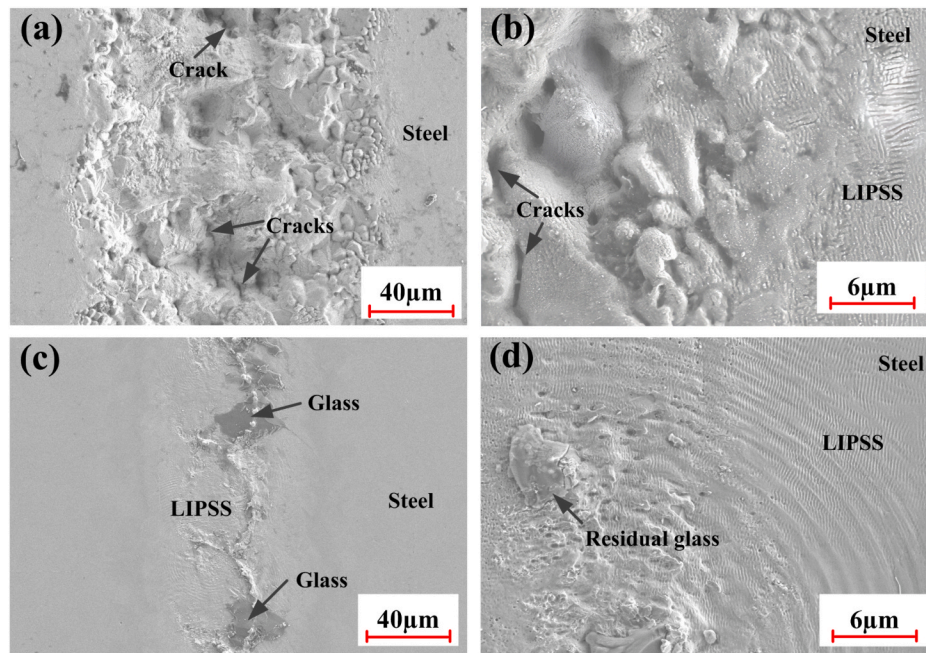


Fig. 3. Microstructure of steel-glass weld interface in the absence of pressure (a-b) and with pressure assistance (c-d): (a) low magnification at 0 N; (b) high magnification at 0 N; (c) low magnification at 400 N; (d) high magnification at 400 N.

3. Results and discussion

3.1. Interfacial microstructure

3.1.1. Microstructure under different process conditions

In order to study the influence of loading conditions on joint formation, contrastive ultrafast laser welding tests were performed on dissimilar materials of steel and glass. The microstructure of the welded joints under different load conditions is shown in Fig. 3.

As shown in Fig. 3a and b, irregular structures were formed at the steel interface after green picosecond laser welding. The formation of this typical morphology was closely linked to the interaction between picosecond laser and experimental materials. When a picosecond laser pulse strikes, the borosilicate glass shows a nonlinear energy absorption while 304 stainless steel exhibits a linear energy absorption, causing a battery of processes such as melting, ejection of matter, ablation and solidification, which can be used for joining two different materials. During the laser welding process, the molten materials in the irradiated zone occur to migrate and fill the welding gap between them due to the Marangoni effect [29] and material eruption [30], subsequently leading to the formation of heterogeneous joints. Under stress-free conditions, significant ablation and material loss occurred at the interface. In the absence of strong spatial constraints induced by external stress, the recoil pressure from metal vaporization ejected high-energy material. This phenomenon led to expansion of the affected interfacial area, direct laser irradiation on adjacent metal regions increased, causing excessive metal melting and ablation (causing thermal cracks as shown). The high-energy material migrated primarily parallel to the interface under weak spatial constraints (provided only by sample inertia), with only a minimal fraction of molten material contributing beneficially to joint formation through gap filling. It has been also found that some cracks were produced at the weld interface during picosecond laser welding without pressure on the sample surface. The cracks were mainly caused by the thermal stresses formed during the laser welding process. The significant differences in the structures resulted in the material having a poor deformation capacity. The thermal stresses formed in the joints can fail to be completely neutralized by the plastic deformation of the material itself during the laser welding process, subsequently leading to the

occurrence of the cracks.

The wavy structures at the interface can be clearly observed in Fig. 3c and d during picosecond laser welding of glass and steel with the assistance of pressure. These typical structures are called laser-induced periodic surface structures (referred to LIPSS, ripples) [31,32], which are mainly formed by the interaction between the laser with the metal surface. When the laser irradiates material surface, a standing wave field is formed on account of the interference between the scattered wave and the incident laser, thereby resulting in a periodic electromagnetic field distribution and the formation of surface plasmon polaritons (SPP). The SPP interferes with the incident laser to further enhance the periodic electromagnetic field, which leads to quasi-periodic energy absorption on the surface of the material and the emergence of periodic peak-valley structures. The presence of these structures at the interface effectively increases the bonding areas, thereby enhancing the binding force between the glass and the stainless steel. Based on this reason, the following study was carried out by pressure assistance.

3.1.2. Microstructure under different process parameters

To further explore the microstructure evolution at the interface, this study conducted welding experiments on glass-to-steel dissimilar materials under two parameter sets: (1) process tests with repetition frequency of 800 kHz, welding speed of 50 mm/s and different laser powers, and (2) process tests with laser power of 16W, welding speed of 50 mm/s and different repetition frequencies. The corresponding microstructural evolution characteristics are presented in Figs. 4 and 5, respectively.

It can be clearly observed from Fig. 4 that disparate microstructures at the interface can be obtained under the action of different laser powers. When the glass and steel were welded with low laser power, the uneven weld was formed in Fig. 4a. After shear removal of the glass above, only minimal glass residues remained as block products embedded in the metal matrix in Fig. 4b. These residual glass regions constituted the actual effective bonding zones in the joints. The laser failed to deliver sufficient energy to establish larger effective bonding areas. With the increase of picosecond laser power, the granular glass products at the interface were produced instead of block glass products in Fig. 4c. Further observation found that the wavy structures were also

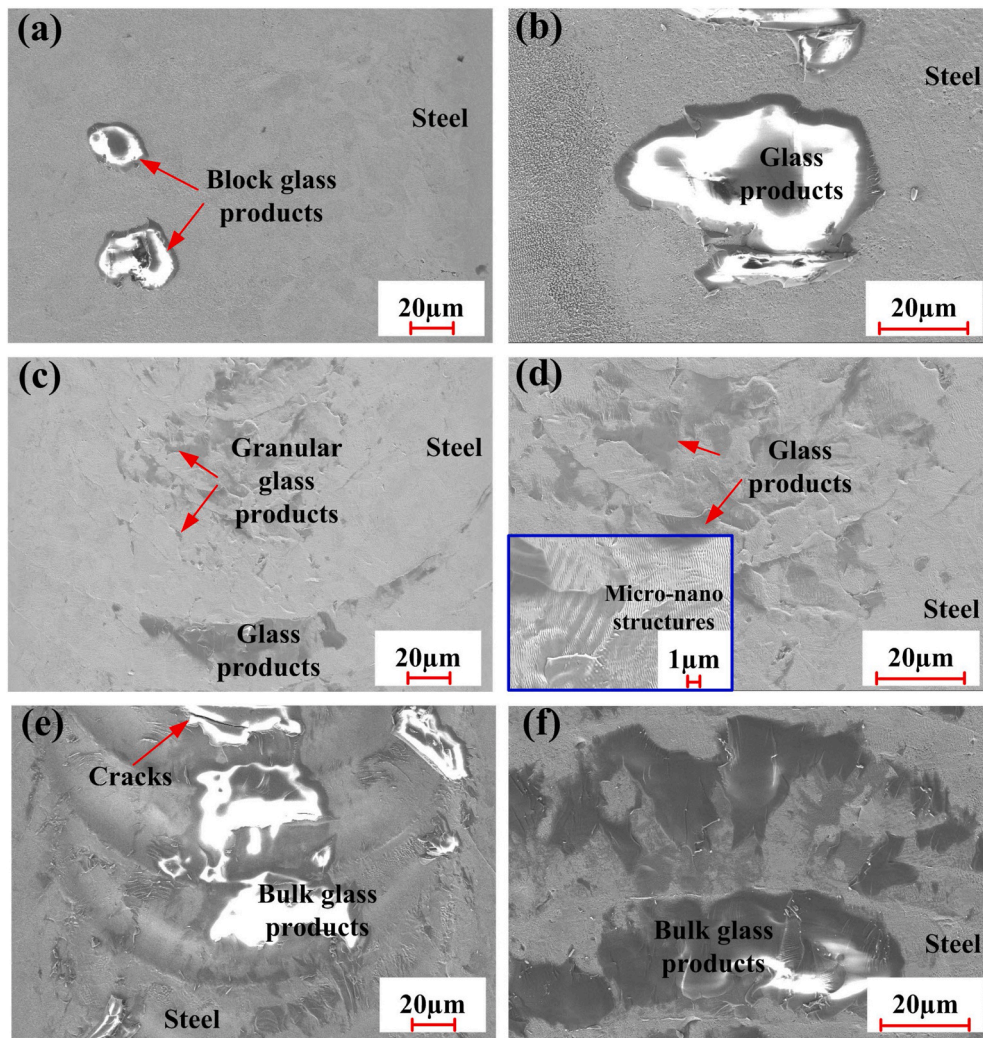


Fig. 4. Microstructure of steel-glass weld interface at different laser powers: (a) low magnification and (b) high magnification at 12 W; (c) low magnification and (d) high magnification at 16 W; (e) low magnification and (f) high magnification at 20 W.

formed among glass products in Fig. 4d. Refinement of precipitates and formation of laser-induced periodic surface structures can effectively improve interface bonding property.

However, with the further increase of picosecond laser power, more and more glass in the irradiation zones was melted into the joint, causing much more blocky glass products were produced at the interface after solidification in Fig. 4e and f. The enrichment and association of these solidified products can increase the crack sensitivity because of their great brittleness. The cracks easily occurred at the interface of glass and steel under the action of welding thermal stress, thereby resulting in a reduction in joint performance.

Fig. 5 show the interface microstructure of glass-steel joints welded with laser power of 16 W, welding speed of 50 mm/s and different repetition frequencies. As demonstrated in Fig. 5a, it can be observed that local glass solidification products were enriched and remained at the interface when picosecond pulse laser welding with low repetition frequency. Moreover, LIPSS were formed near the glass products in Fig. 5b. The difference of LIPSS can directly affect the binding effect at the interface of glass and steel. When the repetition frequency increased to 800 kHz, the molten glass was distributed discretely at the interface, subsequently forming a strong bonding surface in Fig. 5c. It has been found in Fig. 5d that small LIPSS were produced near the glass products. The presence of LIPSS with submicron-scale periodicity increases the bonding surface area between glass and metal while maintaining the

discrete distribution pattern of the glass phase. It was very beneficial to enhance the bonding force between dissimilar materials, since this combination of increased surface area and morphological interlocking is highly advantageous for enhancing the adhesion between these dissimilar materials.

The increase in repetition frequency to 1000 kHz enhanced the non-linear absorption of laser energy within the glass component itself. This localized energy concentration, rather than a general increase in interfacial heating, resulted in a greater volume of molten glass phase at the interface, as evident in Fig. 5e. The increase of brittle products can reduce the deformation ability of the material and lead to the occurrence of micro-cracks. Further research found that large scale LIPSS increased at the interface in Fig. 5f. It is because the heat cumulative effect in the joint brought about a greater temperature rise as the frequency increased during the laser welding process. The change altered the absorption rate of the material, and then affected the morphology of the LIPSS, ultimately resulting in an increase in the period and structural irregularities. The bonding force between the two materials can be weakened because of the increase of solidified glass products and the transformation of microstructure.

3.2. Joint cross-section microstructure at optimized process parameters

In order to further explore the interfacial characteristics in glass-steel

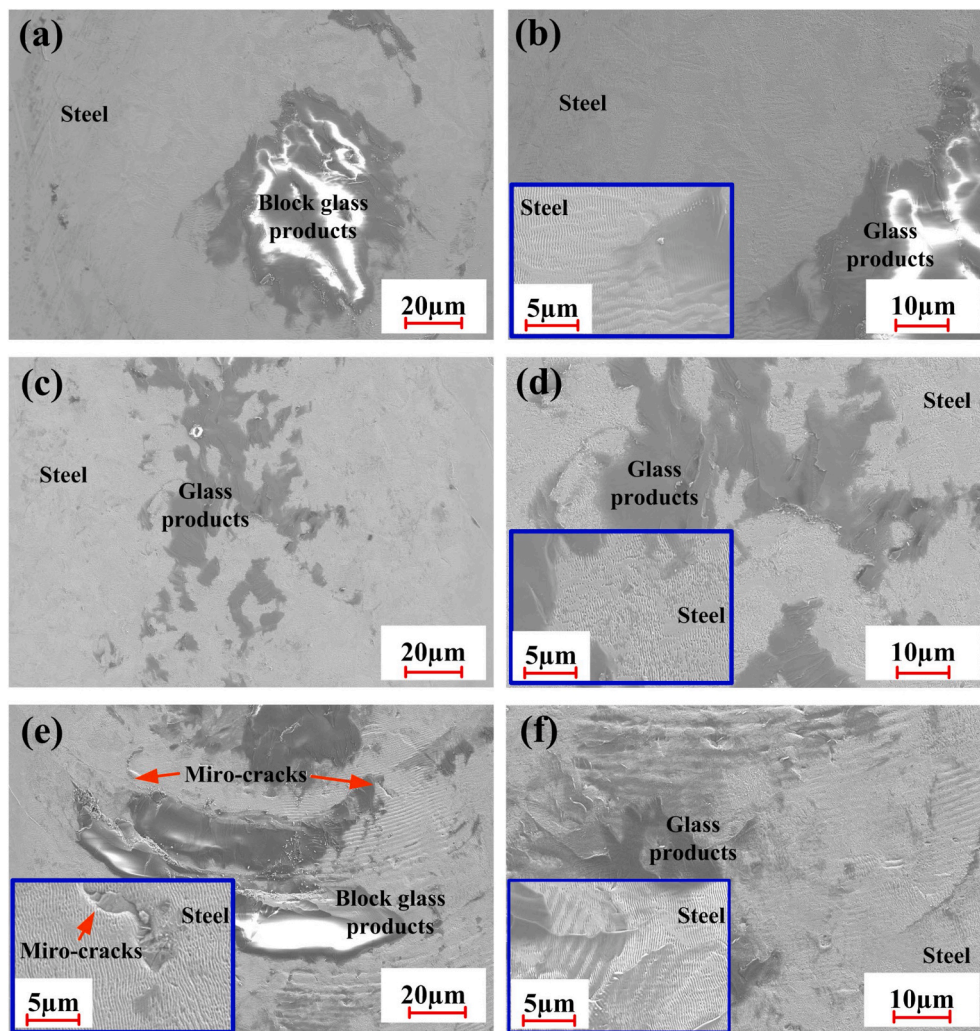


Fig. 5. Microstructure of steel-glass weld interface at different repetition frequencies: (a) low magnification and (b) high magnification at 600 kHz; (c) low magnification and (d) high magnification at 800 kHz; (e) low magnification and (f) high magnification at 1000 kHz.

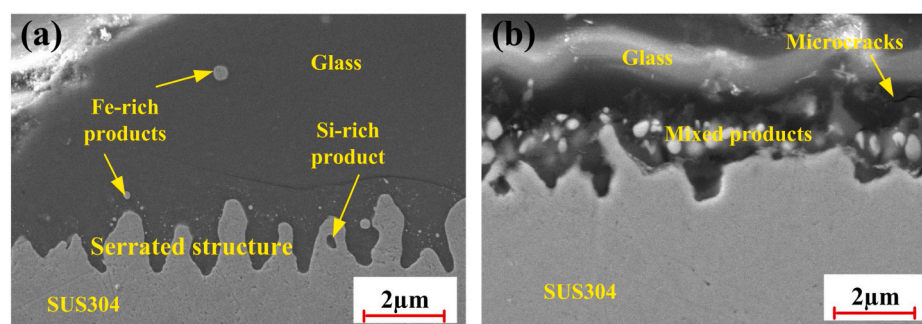


Fig. 6. SEM images showing the cross-sectional microstructure of joints welded at 800 kHz repetition frequency: (a) serrated structure; (b) irregular structure.

joints, a sample was welded with optimum parameters to observe the cross sectional morphology. A cross-sectional view of the glass and stainless steel weld is shown in Fig. 6.

The SEM examination, demonstrated in Fig. 6, revealed that two typical morphologies including were formed at the interface of glass-steel joints. As seen from Fig. 6a, it can be observed that the serrated structures were produced at the stainless steel side during the welding process. These structures were firmly embedded in the glass matrix and played a nailing role in the connection process. It was very beneficial to enhance the junction of glass and stainless steel. It was also found that

globular Fe-rich phases of different sizes were discretely distributed in the solidified glass products while part of Si-rich phases were trapped in the solidified steel products. It can be concluded that a metallurgical reaction occurred, resulting in the recombination of the matter during the welding process.

As shown in Fig. 6b, the irregular structures were detected in the remote areas irradiated by lasers. Mixed products including Fe-rich phases and Si-rich phases were produced at the interface of glass and stainless steel because of their interaction induced by picosecond laser. Moreover, micro-cracks were formed at the junction of glass and

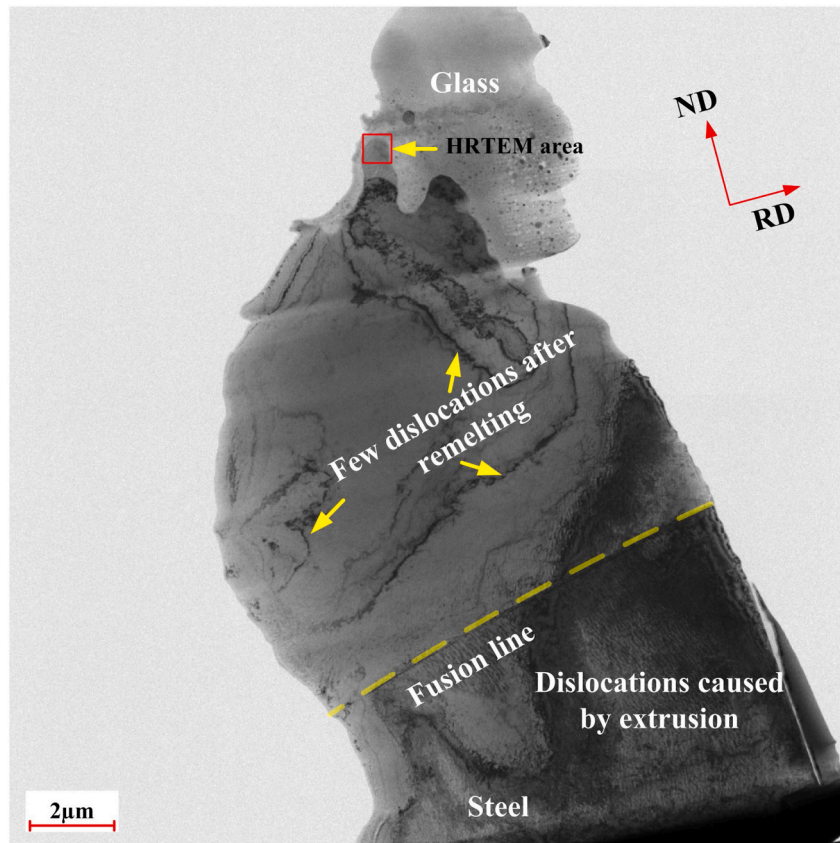


Fig. 7. TEM image showing the microstructure in the interfacial reaction zones.

stainless steel. The occurrence of cracks was mainly caused by the thermal stresses during the laser welding process. Due to the huge difference in physical and chemical properties, large thermal stresses can occur at the interface of glass and stainless steel. The stresses easily fail to be completely eliminated by self-deformation because the glass has poor deformation ability. The brittle binding site is prone to cracking when subjected to force, thereby producing a weak link in the glass and steel joint.

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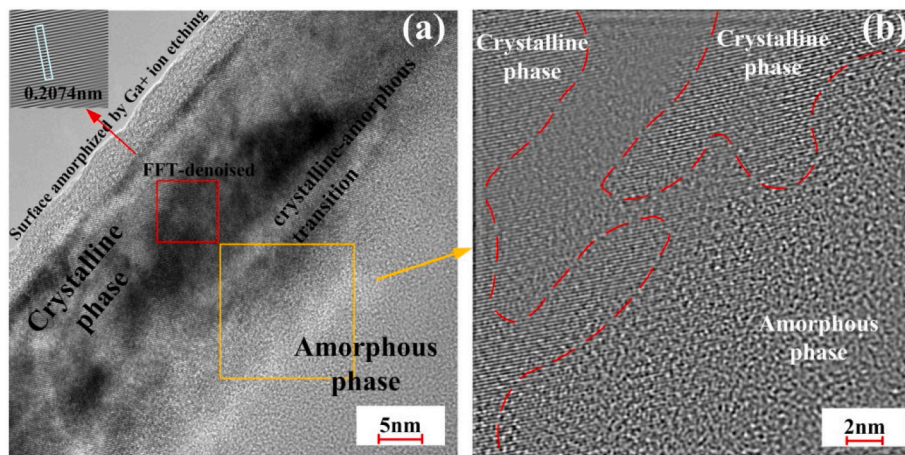


Fig. 8. HRTEM images of the transition zone between crystalline and amorphous phases: (a) microstructure in the transition zone; (b) staggered distribution characteristic in the transition zone.

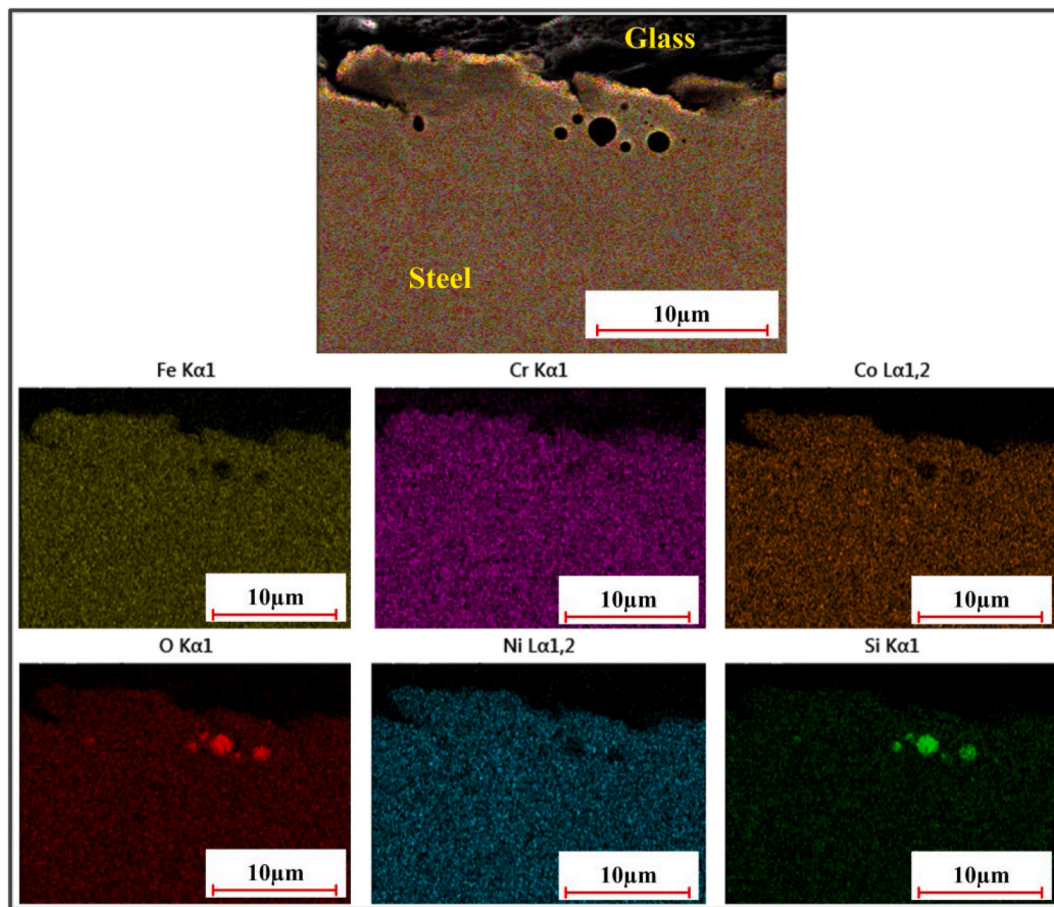


Fig. 9. EDS results of the glass-steel interface of green picosecond laser welded joints.

3.3. Interface structure of laser melting zones

In order to further investigate the interfacial microstructure in laser melting zones, Focused Ion beam (referred to as FIB) was used to fabricate extremely thin glass-metal composite sample. The experimental results of the measured sample are shown in Fig. 7.

As shown in Fig. 7, the fusion line of the joint can be observed on the metal side. The thickness of the molten layer was about 5 μm on the metal side. Due to its amorphous structure, the constructive interference is not obvious when the incident electron beam diffracts. It is characterized by a clear distribution of light and dark on the metal side. The main reason was that the lattice distortion around the dislocation in the metal crystal causes the variation of local diffraction conditions, thereby resulting in the difference in diffraction intensity between the dislocation region and the surrounding matrix. It has been also observed that only a few dislocation lines and dislocation cells existed in the molten layer of the metal while a dislocation network was formed outside the molten zone due to the accumulation of a large number of dislocations. This is because the metal surface heats up rapidly after the free electrons absorb the photon energy, and then the energy is transferred to the lattice through the electron-phonon coupling. The metal material melts at the highest incident energy position. The laser-induced compressive stress relaxation induces the instability of the molten material at the upper part, resulting in the growth of the surface structure and the relief of the internal stress in the melt. However, the laser-induced thermal stress outside the melt zone remains under the constraint of inertial stress, resulting in a large amount of dislocation accumulation in the melt zone and affecting the solidified metal melt.

In order to further study the precipitated phase in the transition zone, high resolution transmission electron microscopy (referred to HRTEM)

was used to observe the microstructure in the area marked by the red box in Fig. 7. This higher-resolution analysis, as demonstrated in Fig. 8, reveals the presence of a transition layer between the crystalline steel and the amorphous glass.

As shown in Fig. 8a, it can be found that the microstructure in the transition zone consisted mainly of amorphous phase, crystal phases, and mixed phases of the two. Among these phases, the crystal phase was identified as the ferritic phase according to the calculation of fringe spacing by fast Fourier transform (referred to as FFT) denoising. The width of the transition curve between the amorphous and crystal phases is about 4 nm. It should be noted that there was an amorphous phase layer of uniform thickness in the upper-left part of the sample. This was due to the ion implantation effect when Ga^+ ions bombard the metal surface manufactured by the FIB method. During this process, Ga^+ ions were injected into the metal material, which destroyed the original crystal structure and led to the amorphous region. Further observation found that the crystalline and amorphous regions had the characteristics of staggered distribution and there was a transition zone between the region with high crystallinity and the region with complete amorphous in Fig. 8b. This heterogeneity of microstructure may allow the material to achieve a balance between strength and toughness.

3.4. Element distribution near the interface

In order to explore the reaction mechanism of the interface during picosecond laser welding, energy spectrum analysis was performed on dissimilar materials of glass and steel. The experimental results of the tested specimen are shown in Fig. 9.

It can be clearly observed in Fig. 9 that Si and O elements occurred in the steel matrix. It can be inferred that a metallurgical reaction has

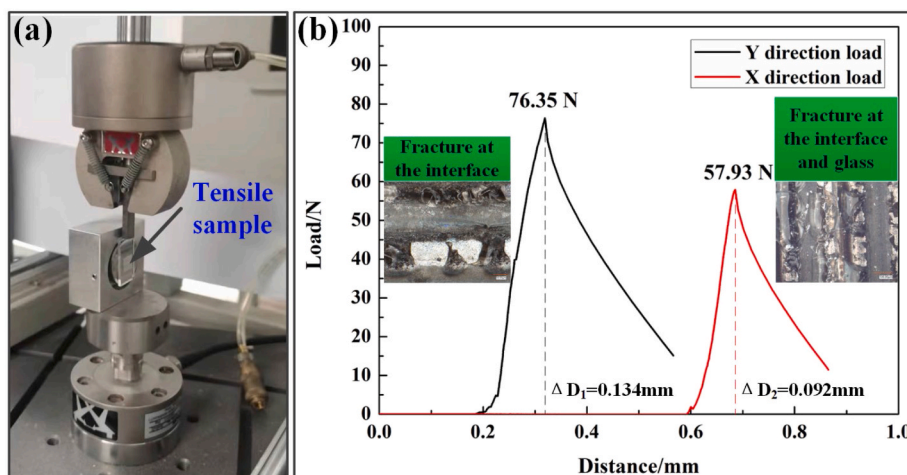


Fig. 10. The mechanical property of laser welded joint: (a) tensile instrument; (b) tensile curves along the different directions.

occurred between glass and steel during the laser welding process. Due to the segregation and enrichment of the elements at the interface, multiple spherical precipitates were produced by means of the movement of the molten pool and the migration of elements. These heterogeneous precipitates were haphazardly distributed in the interface reaction layer. Moreover, some alloying elements such as Cr and Fe, entered into the glass during the welding process because of the intense heat action and the convection in the molten pool. The migration of elements can effectively reduce the composition segregation, thus decreasing the structure stress in the joints to some extent. Due to the difference in thermal expansion coefficients between glass and stainless steel, the existence of these precipitates in clusters was harmful to the junction of dissimilar materials of glass to stainless steel. The precipitated heterogeneous phases may cause the stress concentration or formation of shrinkage-induced crack initiation at the edges, which can easily weak the interfacial bonding force because of its great brittleness.

3.5. Mechanical property of the joint

To evaluate the mechanical property of the laser-welded joints, a shear tensile test was carried out on the samples with optimized parameters. The experimental results of measured specimens are shown in Fig. 10.

As shown in Fig. 10, the tensile forces of glass-steel welded joints along the X direction (welding direction) and Y direction (Perpendicular to the welding direction) were 57.93 N and 76.35 N, respectively. The average value of the tensile forces was up to 67.14 N. Under the action of external forces, the bearing capacity of the joints quickly reached the maximum and then rapidly decreased. It can be inferred that the joint had poor plastic deformation capacity during the stretching process.

From the previous analysis, it can know that interface reaction layers and serrated structures were formed in the interaction between laser and material. The former played a metallurgical bonding role, and the latter played a mechanical bite role. The improvement of joint performance depended mainly on the synergistic action of the two structures. However, the formation of blocky glass products at the interface can seriously weaken the bonding force between glass and steel because of its great brittleness.

4. Conclusions

In this study, direct welding with 170 mm-focus green picosecond laser was conducted on dissimilar materials of borosilicate glass and stainless steel. The interfacial microstructures of laser-welded joints were carefully investigated. The joining mechanism of borosilicate glass

and stainless steel was deeply explored. Moreover, the mechanical property of welded joints was analyzed in detail. The main conclusions are summarized as follows:

- (1) A joint with good weld formation was successfully achieved using green picosecond laser welding with galvo scanning system. The appropriate process parameters were pressing force of 400 N, pulse laser power of 16 W, repetition frequency of 800 kHz and welding speed of 50 mm/s.
- (2) The interface structures were mainly determined by the pressing force during the welding process. The LIPSS and blocky glass were produced at the interface of borosilicate glass and stainless steel with the assistance of pressing force. The microstructural evolution at the interface was closely associated with the process parameters during the welding process.
- (3) The serrated structures were formed at the interface because of the interaction between the laser and the material, which played a nailing role in the connection process. However, the huge differences of the structures can generate the thermal stresses in the interface layers, subsequently inducing the occurrence of micro-cracks during the laser welding process.
- (4) The phases formed in the transition zones consisted mainly of amorphous phase, crystal phases, and mixed phases of the two. Compared to pure mechanical interlocking, the transition zone gradually mitigates thermal mismatch stress through continuous variations in composition and structure, thereby reducing stress concentration and enhancing interfacial bonding strength.
- (5) The improvement of the joint performance was mainly attributed to the pinning effect of the serrated structures at the interface and the reduction of microstructural gradients in the interface reaction layers.

Declaration of competing interest

The authors do not have any possible conflicts of interest.

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